ROTATION AND REVOLUTION

1. Introduction to Earth's Movements 🌍🏞️

Key Points:

- The earth revolves around the sun in an elliptical orbit and also rotates on its axis.
- These two motions are known as *revolution* and *rotation*, respectively.
- Many phenomena on earth like day and night, seasons, winds, and ocean currents depend on these motions.

What does it basically mean? 🤔 Our planet never sits still; it spins like a top while running laps around the sun, and these two movements are the reason we have different weather, days, and seasons! 🏃‍♂️🌞

2. Rotation of the Earth 🌪️📍

Definitions:

- **Rotation:** The spinning of the earth on its axis from west to east once in 24 hours.
- **Axis:** The imaginary line passing through the centre of the earth as it does on the globe.

Key Points:

- The two endpoints through which the axis passes are known as the North Pole and the South Pole.
- The axis is tilted at an angle of $23\frac{1}{2}°$ from the vertical.
- If a line were drawn parallel to the plane of the earth's orbit, the axis would make an angle of $66\frac{1}{2}°$ to this line.

What does it basically mean? 🤔 The Earth doesn't stand perfectly straight up and down; it is permanently tilted sideways as it spins, creating an imaginary pole right through its center. 💈📐

3. Characteristics of Rotation ⚙️⏱️

Key Points:

- **1. Direction:** The earth completes one rotation from west to east on its axis in 23 hours, 56 minutes, 4.09 seconds (which is roughly 24 hours).
- **2. Inclination of Axis:** The earth is always tilted in the same direction (towards the Pole Star) as it moves round the sun. The angle at which the earth is tilted is known as the *inclination of the earth's axis*.
- **3. Speed:** The speed of earth's rotation at the Equator is 1670 km per hour (because the circumference of the earth is 40,075 km, which is covered in 24 hours). The speed decreases towards the poles.

What does it basically mean? 😳 The Earth is a giant, fast-moving vehicle spinning eastward, where the middle (Equator) spins the fastest, while the top and bottom (poles) barely move at all! 🚗💨

4. Effects of the Rotation of the Earth 🌅🌊

Definitions:

- **Dawn:** The time when the sun begins to cast its light in the sky.
- **Noon:** The time when the sun is overhead.
- **Dusk:** It is twilight and the sun is seen disappearing in the sky.
- **Night:** At night, it is completely dark.
- **Coriolis Effect:** The force experienced by any object moving over the surface of a rotating body such as the earth.
- **Centrifugal force:** An apparent outward force that acts on a rotating object.

Key Points:

- **1. Occurrence of Day and Night:** Caused by the rotation of the earth on its axis. Parts of the earth which face the sun have *day* and parts which do not face the sun have *night*. This happens with precision and progression and not suddenly.
- **Duration:** The duration of the day and night is not equal at all places on the earth because of the inclined axis. The length of days varies with respect to the seasons as well as latitude.

2. Sunrise and Sunset: Caused by rotation from west to east. Every part of the surface faces the sun at some time. The part moving from darkness into the sun's rays experiences *sunrise*, and when moving away from the rays, it experiences *sunset*.

- **3. Coriolis Effect (Winds & Oceans):** Plays an important role in the movement of air (winds) and water (ocean currents). Due to this effect, winds and ocean currents move clockwise and are deflected to the right in the Northern Hemisphere. They move in the anticlockwise direction and are deflected to the left in the Southern Hemisphere.
- **4. Celestial Movement:** The sun, the moon and the stars seem to move from east to west because the earth spins from west to east. This is similar to experiencing trees moving while looking from a moving train.
- **5. Centrifugal force (Earth's Shape):** It is always directed outward, away from the axis of rotation. The speed of rotation has created a centrifugal force resulting in a bulge in the middle portion of the earth and flattened top at the poles.
- **6. Difference in time:** Rotation causes difference in time over various places on the earth. The difference in time is 4 minutes for each degree of longitude [24 hours (1440 minutes) divided by 360°].
- **7. Heat Gain and Loss:** The side of the earth towards the sun constantly gains heat, while the side away from the sun constantly loses heat by radiating it into outer space.
    - *(i)* The short rotational period makes the variation of temperature between day and night moderate.
    - *(ii)* Rotation in combination with Revolution causes seasons and variations in the lengths of day and night over different places on earth.

What does it basically mean? 🤔 The Earth's spin acts like a giant natural clock and climate controller! It gives us sunrises, shapes the planet like a squished ball, bends our wind paths, and makes sure one side doesn't bake while the other freezes. 🕰️🌡️

1. Effect of the Inclined Axis 🌍📐

Key Points (Exhaustive Detail):

- The duration of day and night is not equal at all places on earth.
- The specific time of the day at any place is determined exclusively by the height of the sun in the sky.
- If the axis of the earth was vertical instead of inclined, places on earth would have the exact same temperature throughout the entire year.
- If the earth did not revolve around the sun, even with its inclined axis, there would be absolutely no seasons.
- *Handwritten Diagram Mastery:* You drew a brilliant mini-diagram of the earth and sun to prove this, labeling the closer distance as **"more temp"** and the further curve as **"less"**! You also noted **"- radii"** at the top of the page.

The "Why?" Section (Mechanics of Seasons) ❓

- **Why do we have changing seasons and varying day lengths?** The path of the earth around the sun is elliptic (passing through the centre of both bodies). The earth's axis is tilted at an angle of $66\frac{1}{2}^{\circ}$ to the plane of the elliptic (or inclined at $23\frac{1}{2}^{\circ}$ from the perpendicular). As a result, the North Pole remains inclined towards the sun for part of the year while the South Pole tilts away, and then the situation completely reverses.
- *(Page-Shift Protocol Engaged!* 🧩*)* If the axis were perfectly perpendicular, there would be equal days and nights at all places, at all times of the year, and there would be no seasons at all!.

2. Why We Do Not Feel the Motions of the Earth 🛑🌍

Key Points:

- Rotation and revolution are the two chief motions of the earth.
- If we were to actively feel these massive motions, it would make life incredibly uncomfortable.
- The powerful force of gravity keeps everything firmly attached to the earth.

- Because of this, gravity keeps all objects completely *at rest* with reference to the motions of the earth.

The "Why?" Section ❓

- **How did scientists prove the earth rotates if we can't feel it?** They used the principle of gravity combined perfectly with Newton's First Law of Motion (which states that a body remains at rest or in uniform motion unless compelled to change that state by an external force) to scientifically prove the earth's rotation!.

3. Revolution of the Earth 🔁🌞

Definitions:

- **Revolution:** The motion of the earth revolving round the sun in an elliptical orbit, which combined with rotation makes the earth's environment complex.
- **Solar Year:** The exact period taken to make one complete round of the sun: 365 days, 5 hours, 48 minutes, and 45.51 seconds.
- **Leap Year:** A year with 366 days. For convenience, a solar year is taken as 365 days and 6 hours; those remaining 6 hours are adjusted every 4 years to add an extra day to February.

Key Points (Characteristics of Revolution):

- The earth moves in its elliptical orbit at a staggering average speed of 29.8 kilometres per second (about 100,000 kilometres per hour).
- The speed of revolution is **not uniform**! When the earth is closer to the sun, the intense gravitational pull makes the earth move much faster than when it is away from the sun.
- The earth's axis is always inclined to its orbital plane at an angle of $66\frac{1}{2}^{\circ}$.
- The motion of the earth's revolution is from west to east (which is the exact same direction as its rotation).

4. Effects of Revolution (Seasons & Orbit) ⛈️📏

Key Points (Seasonal Changes):

- The amount of heat a place receives strictly depends on the exact angle at which the sun's rays reach it.
- When a place receives *vertical/direct rays*, the heat is concentrated on a small area, making it very hot.
- When a place receives *oblique rays*, the exact same amount of heat is spread over a much larger area, diluting the heat.
- *Handwritten Mastery:* You drew a large **"X"** over the text detailing how revolution causes an annual cycle of different seasons, perhaps to emphasize that the tilt is the true primary factor!.
- *Handwritten Mastery:* Under "Inclination of axis and seasonal changes," you boldly crossed out the word "abrupt" and wrote **"Slowly"** to show that because of the tilt, seasonal changes transition gradually rather than abruptly! ✍️⏳ .

Definitions (Perihelion & Aphelion):

- **Perihelion:** The position when the earth is closest to the sun (147.3 million km away).
- **Aphelion:** The position when the earth is farthest from the sun (152 million km away).

Key Points (Positions & Timing):

- Because the earth's path is elliptical, the distance between the earth and the sun constantly varies.
- The minimum distance (Perihelion) is reached on or about **January 3rd** every year.
- *(Page-Shift Protocol Engaged across columns! 🧩)* It is winter in the Northern Hemisphere during Perihelion because the axis is tilted away. While the tilt makes the North cool in winter, its comparative nearness to the sun affects the Southern Hemisphere. Conversely, on about **July 3/4**, the earth reaches Aphelion (farthest from the sun).
- The shape of the earth and its revolution around the sun create distinct heat/temperature zones on the surface (The Torrid Zone, The

Temperate Zone, and The Frigid Zone) based on the angle of the sun's rays.

4. Effects of Revolution (Continued) ⛈️📏

Key Points (Perihelion & Aphelion Dates):

- *(Page-Shift Protocol Engaged!* 🧩*)* The comparative nearness to the sun helps to make the Southern Hemisphere warm in summer.
- On **January 3**, the earth in its orbit is closest to the sun, a time known as **Perihelion**.
- On about **July 4**, the earth is farthest from the sun, and at this time the earth is said to be in **Aphelion**.

5. Creation of Heat Zones 🌡️🌎

Key Points (The Mechanics):

- The spherical shape of the earth, along with its movements around the sun, causes a major difference in the exact angles at which the sun's rays fall on the earth's surface.
- This difference in angles directly causes a difference in the distribution of heat on the earth's surface.
- As a direct result of this uneven heat distribution, the world has been divided into three distinct heat or temperature zones.

The Three Distinct Heat Zones:

- 🔥 **(a) The Torrid Zone:**
  - *Location:* This zone stretches exactly between $23\frac{1}{2}^{\circ}$ N and S latitudes.
  - *Sunlight & Climate:* It more or less receives the direct, vertical rays of the sun. As a result, this entire region experiences a moderately warm climate throughout the year.
- 🌤️ **(b) The Temperate Zone:**
  - *Location:* Sandwiched in between the Torrid and the Frigid zones (i.e., between $23\frac{1}{2}^{\circ}$ and $66\frac{1}{2}^{\circ}$ North and South).

    - *Sunlight & Climate:* In this zone, the sun's rays fall obliquely (at a slant), and hence the region experiences a moderate climate.

- ❄️ **(c) The Frigid Zone:**
    - *Location:* This zone extends from $66\frac{1}{2}^{\circ}$ N to 90° N (or North Pole) and $66\frac{1}{2}^{\circ}$ S to 90° S (or the South Pole).
    - *Sunlight & Climate:* Since the sun's rays fall *extremely obliquely* in this region, it experiences an extremely cold climate.

8. Solstices and Equinoxes 🌓🌞

Definitions:

- **Circle of illumination:** The lighted part of the earth that moves with the apparent movement of the sun, ensuring that only half of the earth faces the sun at any given time. 🌓

Key Points:

- **Summer Solstice (June 21):** The North Pole tilts towards the sun, placing the sun directly overhead at the Tropic of Cancer. The Southern Hemisphere tilts away and experiences winter. 🌞
- **Winter Solstice (December 22):** The Southern Hemisphere tilts towards the sun, while the Northern Hemisphere tilts away. This brings winter to the Northern Hemisphere and summer to the Southern Hemisphere. ❄️
- **Equinox:** On two days a year, the sun shines vertically over the Equator at a 90° angle. This angle decreases as you move away from the Equator. 🎯
- **Equinox Dates:** These two days are March 21 (Spring or Vernal Equinox) and September 23 (Autumnal Equinox). 🌸🍂
- At the Equinoxes, the days and nights are of exactly equal length throughout the entire world. ⚖️
- On March 21, it is spring in the Northern Hemisphere and autumn in the Southern Hemisphere. On September 23, the position is perfectly reversed. 🔁

- As the apparent northward movement of the sun continues from the Equinox up to June 21st (Summer Solstice), variations in the duration of daylight start occurring. 📈

Table of Differences: Solstice vs. Equinox ⚖️

Feature	Solstice	Equinox
Meaning	The sun reaches its highest or lowest point in the sky at noon, resulting in the shortest/longest days.	The sun crosses the plane of the earth's equator, making day and night of equal length.
Word Origin	Latin: 'Sol' (sun) + 'sistere' (to stand still).	Latin: 'aequus' (equal) + 'nox' (night).
Dates	June 21 (Summer) & December 22 (Winter).	March 21 (Spring) & September 23 (Autumn).
Season timing	Happens during summer and winter.	Happens at the start of spring and autumn.

🌟 **SPECIAL INSERT: The 3D Geometry of Equinoxes & Seasons** 🌟

Trying to visualize the 3D geometry of the solar system in your head can be mind-bending! Let's completely demystify this exact thing right now:

- **The Permanent Lean:** The Earth doesn't wobble back and forth; it has a permanent 23.5° tilt, and the North Pole is *always* locked in pointing at the North Star. 🌟
- **The Bridge (Sept 23):** Think of an Equinox not as a destination, but as a bridge. On September 23rd, the Earth is positioned exactly sideways to the Sun. The Sun's rays hit the Equator dead center at a 90° angle. 🌉
- **Why is it Autumn?** It is all about the *direction of the transition*. In the Northern Hemisphere, you are pulling away from Summer heat and heading toward Winter. This cool-down transition phase is what we define

as Autumn. Down South, they are tilting *toward* the sun, making it Spring! 🍁

- **Campfire Visual:** In June, you lean into the fire (Summer). In December, you lean away (Winter). On September 23rd, you are sitting perfectly straight up, but pulling yourself backward. That cooling motion is "Autumn." 🔥🏕️

What does it basically mean? 🧐

Because the Earth is permanently tilted while it orbits, the sun's direct light slowly moves up and down our planet throughout the year. When your half of the Earth leans closest to the sun, you get a Solstice with the longest, hottest day of the year. When the Earth is perfectly sideways to the sun, you get an Equinox. During an Equinox, the sun hits the Equator dead-on, meaning everyone on the planet gets exactly 12 hours of sunlight and 12 hours of darkness! ⚖️🌍

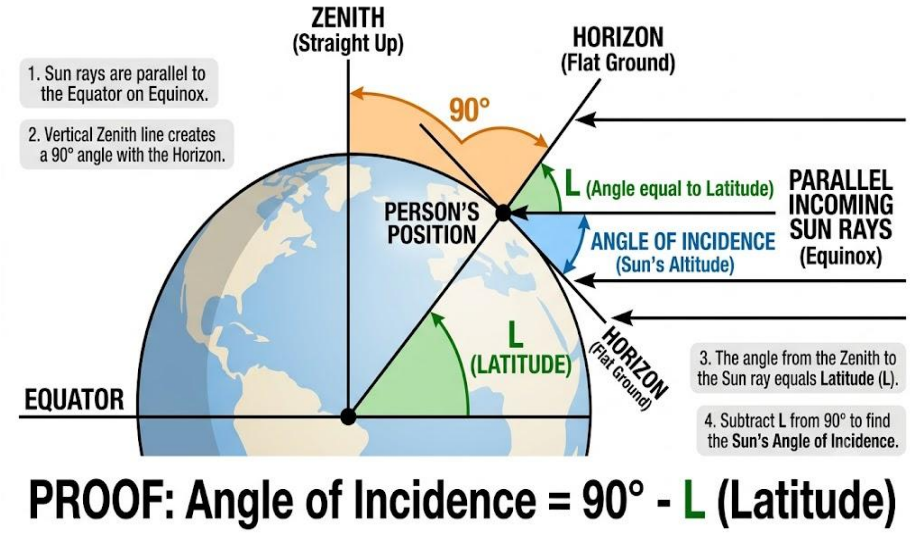
9. Angle of Incidence and Duration of Daylight 📐🌞

Definitions:

- **Angle of Incidence (Altitude):** The angle at which the sun's rays hit the earth, calculated by subtracting the latitude in degrees from 90°. 📏

Key Points:

- The altitude of the sun decreases as you move away from the Equator. 📉
- Seasonal contrasts in daylight and temperature are maximized at mid-latitudes compared to the Equator or the Poles. 🌡️
- Formula: 90° - Latitude = Angle of Incidence. 🧮



Data Tables: Variations by Latitude 📊

Part 1: The Equinoxes (Sept 23 / March 21) 🌸🍂

Latitude	Duration of Daylight	Angle of Incidence
90°N	12 hours	0°
66°30'N	12 hours	23°30'
50°N	12 hours	40°
23°30'N	12 hours	66°30'
0° (Equator)	12 hours	90°
23°30'S	12 hours	66°30'
50°S	12 hours	40°
66°30'S	12 hours	23°30'

Latitude	Duration of Daylight	Angle of Incidence
90°S	12 hours	0°

Part 2: Summer Solstice (June 21) 🌞🥵

Latitude	Duration of Daylight	Angle of Incidence
90°N	24 hours	23°30'
66°30'N	24 hours	47°
50°N	16h 18m	63°30'
23°30'N	13h 27m	90°
0°	12 hours	66°30'
23°30'S	10h 33m	43°
50°S	7h 42m	16°30'
66°30'S	0 hours	0°
90°S	0 hours	0°

Part 3: Winter Solstice (December 22) ❄️🥶

Latitude	Duration of Daylight	Angle of Incidence
90°N	0 hours	0°
66°30'N	0 hours	0°

Latitude	Duration of Daylight	Angle of Incidence
50°N	7h 42m	16°30'
23°30'N	10h 33m	43°
0°	12 hours	63°30'
23°30'S	13h 27m	90°
50°S	16h 18m	63°30'
66°30'S	24 hours	47°
90°S	24 hours	23°30'

What does it basically mean? 🤔

The closer you live to the Equator, the less your daily sunlight changes—it is pretty much 12 hours of day and 12 hours of night all year long. But the closer you move to the North or South Poles, the crazier the changes get! That is why the North Pole has constant daylight for 6 months in the summer, and total darkness for 6 months in the winter. The math is simple: a steep angle of sunlight means weaker heat and wilder daylight hours! 📉❄️

10. Cycle of Seasons 🌸❄️

Key Points:

- The Equinoxes and the Solstices are the positions of the earth with respect to the sun on given fixed dates. 🗓️
- They help us understand how seasons occur and how they are reversed in the Northern and Southern Hemispheres. 🔁
- Because of the continuous variations in the altitude of the sun, the duration of day and night varies according to the seasons. ⏱️

- As a result of these variations, seasons are not the same in all parts of the earth. 🌍
- In general, there are four distinct seasons corresponding to the dates of solstices and equinoxes: Spring, Summer, Autumn, and Winter. 🌸🌞🍂❄️

What does it basically mean? 🧐

The cycle of seasons is just the Earth completing its yearly lap around the sun while permanently leaning to one side. The four major dates (the two Equinoxes and two Solstices) act as the official finish lines dividing up the race track into four equal chunks: Spring, Summer, Autumn, and Winter! 🏎️🏁

11. Key Positions of the Earth During Revolution 🗓️🌍

Definitions:

- **Midnight Sun:** A phenomenon where regions beyond the Arctic or Antarctic Circles experience 24 hours of continuous daylight, with the sun visible at a very low height just above the horizon. 🌞

Key Points:

- **Position on 21st June (Summer Solstice):**
  - It is summer solstice in the Northern Hemisphere as sun rays fall vertically on the Tropic of Cancer.
  - The duration of sunlight increases as you move North: 12 hours at the Equator, **13 hours and 27 minutes at 23½° N**, and **24 hours at 66½° N.** 📈
  - Beyond the Arctic Circle, regions experience 24 hours of daylight (Midnight Sun). This continues for almost three months up to the autumn equinox.
  - Tourists heavily flock to Norway (called the 'land of midnight sun') to witness this. NO
  - In the Southern Hemisphere, it is winter.

    - After June 21st, sunlight hours in the North begin to decrease from 12 hours at the equator to all nights beyond the Arctic Circle until the maximum position is reached on December 21st.

- **Position on 23rd September (Autumnal Equinox):**
    - The sun is vertically overhead at the Equator.
    - Days and nights are equal, meaning the duration of sunlight is exactly 12 hours on all places on earth. ⚖️
    - It is autumn in the Northern Hemisphere and spring in the Southern Hemisphere. 🍂🌸
- **Position on 22nd December (Winter Solstice):**
    - The Southern Hemisphere is tilted towards the sun, experiencing summer, while the Northern Hemisphere experiences winter. ❄️
    - Areas beyond the Antarctic Circle experience 24 hours of daylight. Because it is a water Hemisphere, the spectacle of the Midnight Sun here is confined mostly to navigators. 🚢
    - The South Polar region has 24 hours of continuous sunlight for many days.
- **Position on 21st March (Vernal Equinox):**
    - After Dec 22nd, the sun moves back towards the Equator, falling vertically over it on March 21st.
    - All places on earth have equal days and nights again (a repeat of the Sept 23rd daylight duration).
    - Seasons are reversed: it is spring in the Northern Hemisphere and autumn in the Southern Hemisphere. 🌱🍁

What does it basically mean? 🧐

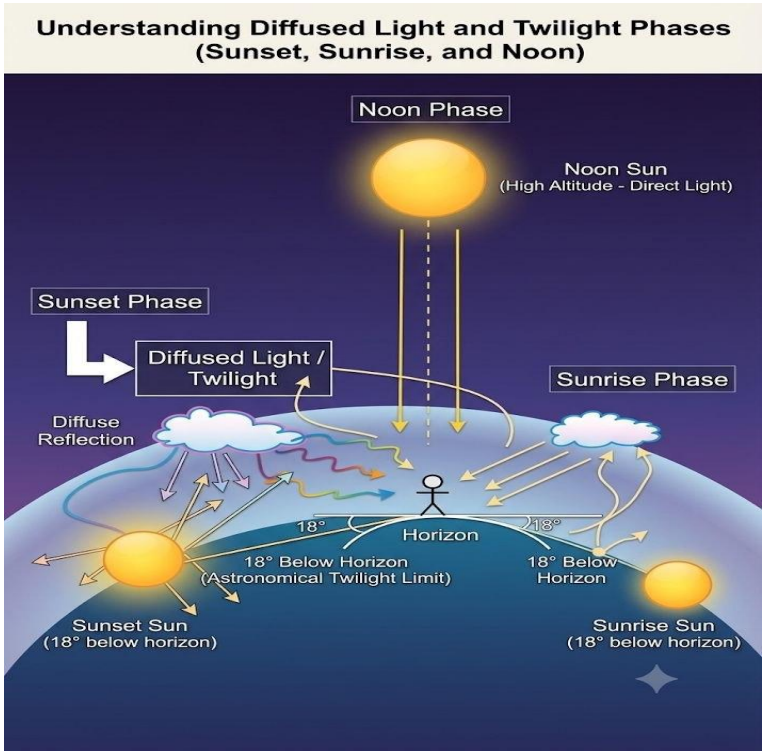
As the Earth drives around the sun, its permanent tilt changes which half gets blasted with direct sunlight. In June, the Northern half gets so much sun that places near the North Pole (like Norway) don't see the sunset for months! By December, the South Pole gets its turn for constant sunlight, though mostly only sailors are there to see it. In March and September, the sun aims perfectly

at the middle (Equator), giving everyone on the planet exactly 12 hours of day and 12 hours of night. ☀️ ❄️ ⚖️

12. Twilight in Low and High Latitudes 🌅🌎

Definitions:

- **Twilight (Dusk):** The period of diffused light between sunset and complete darkness. 🌇
- **Dawn:** The period of diffused light between night and full daylight (sunrise). 🌅
- **Diffused Light:** When the sun is physically below the horizon but we still experience light because its rays are reflected by clouds and dust particles present in the atmosphere. ☁️



Key Book Points (Combined & Verbal Notes):

- There are great seasonal contrasts as you move from the Equator to the Poles because the amount of heat received depends entirely on the inclination (angle) of the sun's rays. 🌡️
- There is no abrupt change from vertical to slanting rays anywhere on earth; the change is gradual.
- The phenomena of dawn and dusk are experienced much more in temperate regions.
- Twilight scientifically continues as long as the sun is **18° below the horizon**. 📐

- The duration of twilight has a considerable influence on the seasons.
- **At the Equator (Low Latitudes):**
    - The duration of twilight is extremely short, specifically **1 hour and 12 minutes**. ⏱️
    - *Why so short?* The speed of the earth's rotation at the equator is very high. Because of this high speed, and because the sun rises and sets in a steep, vertical (perpendicular) path, the period where refracted light can be received is very quick. ⬇️
- **At the Poles (High Latitudes):**
    - The duration of twilight is large beyond 50° latitude. In polar regions, it lasts for as much as **seven weeks at a stretch**. At the actual poles, the duration is **50 days long** during the 3 months of alternating day and night. ⏳
    - *Why so long?* At higher latitudes, a specific point on the earth has to cover less distance, meaning the rotational speed is much lower. Additionally, the sun rises and sets in a highly oblique (slanting) path. Because of the slower rotational speed and the slanted path, the sun lingers right near the horizon, causing the refracted light to stretch out. Winter darkness at the poles is essentially just one massive, long twilight! 📉

Table of Differences: Twilight Duration 📊

Feature	**Low Latitudes (Equator)**	**High Latitudes (Towards Poles)**
Speed of Earth's Rotation	Very High ⏩	Very Low 🐢
Path of the Sun	Vertical / Perpendicular ⬇️	Oblique / Slanted 📉

Feature	Low Latitudes (Equator)	High Latitudes (Towards Poles)
Duration of Refracted Light	Very Short	Very Long
Exact Twilight Duration	1 hour 12 minutes	Up to 7 weeks / 50 days

What does it basically mean? 🧐

Think of the horizon like a finish line. At the Equator, the Earth is spinning super fast and the sun drops straight down like a rock crossing that line, so the sky goes from daylight to pitch black very quickly (just over an hour). But up near the North and South Poles, the Earth is spinning much slower. Combined with the fact that the sun slowly glides down at a highly slanted angle, it takes forever to cross that 18-degree mark. This keeps the sky glowing with reflected light for up to 50 days! The speed of the spin and the angle of the sun's drop completely control how long twilight lasts. 🏊‍♂️🌅

Exercises

1. **Which of the following is true about the rotation of the earth?**
   - **Answer:** All of the above 🌎
2. **See the above figure. What is the value of angle A?**
   - **Answer:** $23\frac{1}{2}°$ 📐
3. **An apparent outward force that acts on a rotating object:**
   - **Answer:** Centrifugal Force 🌪️
4. **Assertion (A): Rotation is the spinning of the earth on its axis from west to east once in 24 hours. Reason (R): The earth is always tilted in the same direction as it moves round the sun.**
   - **Answer:** Both A and R are true but R does not explain A. ⚖️

5. Assertion (A): The winds and ocean currents move clockwise and are deflected to the right in the Northern Hemisphere. Reason (R): Coriolis Effect refers to the force experienced by any object moving over the surface of a rotating body such as the earth.

- **Answer:** Both A and R are true and R is the correct explanation of A. 🌊

6. What is the shape of the path of the earth around the sun?

- **Answer:** Elliptic 🪐

7. Which of the following is correct?

- **Answer:** Aphelion : earth is farthest from the sun. 🌞

8. What is the time difference between two consecutive lines of longitude?

- **Answer:** 4 minutes ⏱️

9. Which of the following is true about the revolution of the earth around the sun?

- **Answer:** All of the above 🚗

10. Assertion (A): A bulge in the middle portion of the earth and flattened top at the poles. Reason (R): Centrifugal force always directed outward, i.e., away from the axis of rotation.

- **Answer:** Both A and R are true and R is the correct explanation of A. 🌐

11. Assertion (A): In the course of the earth's movement around the sun, the North Pole remains inclined towards the sun for a part of the year while the South Pole is tilted away from it. Reason (R): The earth rotates on an inclined axis in an elliptical path around the sun.

- **Answer:** Both A and R are true and R is the correct explanation of A. 🔥

12. Which of the following is true about the perihelion position of the earth?

- **Answer:** It occurs on 3 January. 🗓️

13. The lighted part of the earth is called the _____

- **Answer:** Circle of Illumination 💡

14. Perihelion : 147.4 million km :: _____ : 152 million km

- **Answer:** Aphelion 🌌

15. Which of the following is true about the aphelion?

- **Answer:** Occurs on 4 July. 🎆

16. Assertion (A): The speed of revolution of the earth is uniform. Reason (R): When the earth is closer to the sun, the gravitational pull makes the earth move faster than when it is away from the sun.

- **Answer:** A is false but R is true. 🚀

17. Solstice : Summer/Winter :: _____ : Spring/Autumn

- **Answer:** Equinox 🌸

18. Name the phenomenon when the regions beyond the Arctic Circle experience 24 hours daylight.

- **Answer:** Midnight sun 🌞

19. The period of diffused light between sunset and complete darkness is called

- **Answer:** Both dusk and twilight 🌅

20. The duration of twilight:

- **Answer:** All of the above 🌠

21. Assertion (A): Phenomena of dawn and dusk is experienced more in temperate regions. Reason (R): The twilight continues as long as sun is 18° below the horizon.

- **Answer:** Both A and R are true but R does not explain A. 📐

II. Short Answer Questions 🗺️

1. What is meant by the inclination of earth's axis? At what angle is the earth's axis inclined?

- **Answer:** The inclination of the Earth's axis refers to the permanent tilt of its imaginary axis of rotation relative to its orbital plane. The Earth's

axis is inclined at an angle of 23½° from the vertical, or 66½° to the orbital plane. 📐🌍

2. **Name the two chief motions of the earth.**

- **Answer:** The two chief motions of the Earth are Rotation (spinning on its axis) and Revolution (orbiting around the Sun). 🔄🏎️

3. **Describe two characteristics of rotation.**

- **Answer:** First, the Earth rotates on its axis from West to East. Second, it takes approximately 24 hours to complete one full rotation, which causes the daily cycle of day and night. 🕰️🌅

4. **Mention any two possible consequences if the axis of the earth was vertical instead of inclined.**

- **Answer:** If the axis were vertical, there would be absolutely no cycle of seasons. Additionally, every single place on Earth would experience exactly 12 hours of day and 12 hours of night all year round. ⚖️🚫🍂

5. **State one reason why we do not feel the motions of the earth.**

- **Answer:** We do not feel the Earth spinning because everything on the planet, including us and the atmosphere, is moving along with it at a constant, steady speed, and gravity keeps us firmly anchored. 🏃‍♂️🌍

6. **What is an elliptic?**

- **Answer:** An elliptic (or elliptical) path is an elongated, oval-shaped orbit. The Earth travels in this elliptical shape around the Sun, meaning its distance from the Sun changes throughout the year. 🪐📏

7. **State two chief characteristics of the revolution of the earth.**

- **Answer:** First, the Earth revolves around the Sun in an elliptical orbit taking about 365¼ days. Second, the Earth's axis remains tilted in the exact same direction throughout this entire journey, which causes the changing seasons. 🌸❄️

8. **What do you understand by perihelion and aphelion?**

- **Answer:** Perihelion is the point in the Earth's orbit when it is closest to the Sun (around January 3). Aphelion is the point when the Earth is farthest from the Sun (around July 4). 🌞📏

9. On which two days are the days and nights equal all over the world and why? What name do you give to these days?

- **Answer:** The days and nights are equal on March 21st and September 23rd. This happens because the Earth is positioned exactly sideways to the Sun, and the vertical rays fall directly on the Equator. These days are called Equinoxes. 🌓⚖️

10. Which is the longest and which is the shortest day in the Northern Hemisphere and why?

- **Answer:** The longest day is June 21 (Summer Solstice) because the Northern Hemisphere is physically leaning towards the Sun. The shortest day is December 21 (Winter Solstice) because the Northern Hemisphere is leaning away from the Sun. 🌞❄️

11. On which dates does the sun shine vertically overhead at (a) Equator, (b) Tropic of Cancer, (c) Tropic of Capricorn?

- **Answer:** (a) Equator: March 21 and September 23. (b) Tropic of Cancer: June 21. (c) Tropic of Capricorn: December 21. 🎯🌍

12. Why is noon hotter than morning?

- **Answer:** Noon is hotter because the Sun is directly overhead, meaning its vertical rays travel a shorter distance through the atmosphere and are intensely concentrated over a small area. In the morning, slanting rays are spread out over a larger area, losing heat. 🥵🌅

13. Why are days and nights equal at all places on earth on September 23? What are the seasons in the Northern and the Southern Hemispheres on this date?

- **Answer:** Days and nights are equal because September 23 is an Equinox, where the Sun's rays fall perfectly vertically on the Equator. On this date, it is Autumn in the Northern Hemisphere and Spring in the Southern Hemisphere. 🍂🌸

14. What is meant by Summer Solstice? When do we have Summer Solstice in the Northern Hemisphere?

- **Answer:** The Summer Solstice is when the Sun reaches its highest apparent point in the sky, resulting in the maximum amount of daylight for the year. In the Northern Hemisphere, this occurs on June 21. 🏖️🌞

15. What is meant by Equinox?

- **Answer:** Equinox translates to "equal night." It is the phenomenon where the Sun's rays hit the Equator directly, resulting in exactly 12 hours of daylight and 12 hours of darkness everywhere on Earth. ⚖️🌓

16. What will be the duration of daylight on March 21st at 23°30'N and 66°30'S latitudes? Give reason.

- **Answer:** The duration of daylight will be exactly 12 hours at both latitudes. March 21st is an Equinox, which means that every single location across the entire globe experiences equal day and night. ⏳⚖️

17. Briefly describe the cycle of seasons with the change in inclination of the sun's rays. Which temperature zone receives almost vertical rays of the sun and which zones receive slanting rays?

- **Answer:** As Earth revolves, its tilted axis causes the Sun's rays to strike at different angles, creating seasons. The Torrid Zone receives almost vertical, intense rays. The Temperate and Frigid Zones receive slanting rays, causing cooler seasons. 🌍🌡️

18. How is the duration of twilight influenced by seasons?

- **Answer:** Twilight lasts longer during the summer than in the winter. In summer, the Sun sets at a shallower angle relative to the horizon, so it takes longer for it to reach the 18° below horizon mark where complete darkness begins. 🌅🌠

19. How has the phenomenon of the 'Midnight Sun' come about?

- **Answer:** The Midnight Sun occurs because of the Earth's 23½° tilt. During summer in polar regions (like the Arctic Circle in June), the North Pole remains permanently tilted towards the Sun, so the sun never dips below the horizon, causing 24 hours of continuous daylight. 🌞🕰️

20. Mention one effect of seasons in low and high latitudes.

- **Answer:** In low latitudes (near the Equator), seasons have very little effect, keeping temperatures and daylight hours mostly constant. In high latitudes (near the poles), seasons cause extreme shifts, from 24-hour daylight in summer to 24-hour darkness in winter. 🌡️❄️

21. Draw a well labelled diagram to show changes in the altitude of the midday sun at different times of the year.

Ans. ASK THE TEACHERRR

22. Your family had gone to London during your summer break. Your sister was surprised that it was daylight at 9:00 pm. How did you explain the phenomenon to her?

- **Answer:** London is located at a high latitude in the Northern Hemisphere. During the summer break (around June), the Northern Hemisphere is tilted heavily towards the Sun, which significantly increases the duration of daylight, causing the sun to set very late in the evening. GB 🌇

23. Your penfriend from New Zealand has come to Agra to welcome the New Year. He is surprised that it is winters in Agra. What did you tell him?

- **Answer:** I would explain that New Zealand is in the Southern Hemisphere, which tilts towards the Sun in December, experiencing Summer. Agra is in the Northern Hemisphere, which is tilted away from the Sun in December, bringing cold Winter temperatures! NZIN ❄️

III. Distinguish between the following 📊

(i) Rotation and Revolution

Feature	**Rotation** 🔄	**Revolution** 🏎️
Definition	The spinning of the Earth on its own axis.	The movement of the Earth around the Sun in its orbit.

Feature	Rotation 🔄	Revolution 🏎️
Time Taken	Takes approximately 24 hours to complete once.	Takes approximately $365\frac{1}{4}$ days to complete once.
Main Effect	Causes the daily cycle of Day and Night.	Causes the annual cycle of Seasons.

(ii) Vertical and Slanting Rays

Feature	Vertical Rays ⬇️	Slanting Rays ↘️
Angle	Strike the Earth straight down (near 90°).	Strike the Earth at a lower, oblique angle.
Heat Intensity	Very intense heat because they concentrate over a smaller area.	Less intense heat because they spread over a larger area.
Atmospheric Travel	Travel a shorter distance through the atmosphere, losing less heat.	Travel a longer distance through the atmosphere, losing more heat.

(iii) Equinox and Solstice

Feature	Equinox ⚖️	Solstice ☀️❄️
Meaning	Means "Equal Night".	Means the Sun stands still at its highest or lowest point.
Sun's Position	Sun's rays fall vertically on the Equator.	Sun's rays fall vertically on the Tropic of Cancer or Capricorn.

Feature	Equinox ⚖️	Solstice ☀️❄️
Daylight Duration	Day and Night are exactly equal (12 hours) everywhere on Earth.	Maximum inequality between day and night lengths (longest or shortest day).

IV. Structured Questions 📝🌎

1. (a) What is meant by rotation of the earth? State three effects of the rotation of the earth.

- **Answer:** Rotation is the spinning of the Earth on its imaginary axis from west to east. 🔄 Three major effects are: 1) It causes the daily cycle of day and night. 🌓 2) It causes the deflection of winds and ocean currents (known as the Coriolis effect). 🍽️🌊 3) It creates the apparent daily motion of the sun, moon, and stars moving from east to west across the sky. 🌌

1. (b) What is the effect of the inclined axis of the earth on day and night?

- **Answer:** The inclined axis causes the length of day and night to vary throughout the year at different latitudes. ⏳ Because of the tilt, days are longer in summer and shorter in winter for any given hemisphere! ☀️❄️

1. (c) Give a geographical reason for each of the following:

- **(i) Kuala Lumpur rotates faster on the earth's axis than London.**
    - **Answer:** Kuala Lumpur is located much closer to the Equator where the Earth is wider (bulging). Because the circumference is larger there, it has to travel a greater distance in the same 24-hour period compared to London, which is at a higher latitude, meaning it must spin faster! 🏃‍♂️🍽️
- **(ii) We always see the sun rising in the east.**
    - **Answer:** The Earth rotates on its axis from West to East. This constant eastward spin makes celestial bodies like the Sun appear

to move in the exact opposite direction, making it look like it rises in the East and sets in the West! 🌅➡️

- **(iii) Norway is called the Land of the Midnight Sun.**
    - **Answer:** Norway is located in high latitudes near the Arctic Circle. During the Northern Hemisphere's summer, the Earth's tilted axis keeps this polar region permanently tilted toward the sun, preventing the sun from ever dipping below the horizon for months at a time! 🌞🕰️

1. (d) Draw a well labelled diagram to show inclination of the earth on its axis.

- **Answer:** *(To draw this 🖍️: Draw a circle for Earth. Draw a dotted vertical line straight through the middle. Then, draw a solid line tilted at $23\frac{1}{2}^\circ$ from that vertical line—this is your axis! Label the top 'North Pole' and bottom 'South Pole'. Draw the Equator perfectly horizontal across the tilted axis.)* 📐🌐

2. (a) Describe how the duration of sunlight changes from the Equator to the Poles with respect to the angle of incidence during the solstices.

- **Answer:** At the Equator, sunlight remains roughly 12 hours year-round. As you move towards the Poles during a solstice, the hemisphere tilting *toward* the sun gets higher angles of incidence (more direct rays) and increasing daylight (up to 24 hours at the pole). The hemisphere tilting *away* gets low, slanting rays and decreasing daylight (down to 0 hours at the pole). 🌞📉

2. (b) Describe how seasons are made and reversed between the Northern and Southern Hemispheres.

- **Answer:** Seasons are caused by the Earth revolving around the Sun while holding a permanent 23.5° tilt. When the Northern Hemisphere tilts toward the sun, it gets direct rays and Summer, while the Southern Hemisphere is tilted away, getting slanting rays and Winter. Six months later, the Earth is on the other side of the Sun, and the tilt causes the exact reverse! 🌸🔄🍂

2. (c) Give a geographical reason for each of the following:

- **(i) The length of day and night is not equal at all places on the earth.**
    - **Answer:** This inequality is entirely caused by the Earth's axis being permanently inclined at 23.5°. Because of this lean, different latitudes are exposed to the sun for different amounts of time as Earth orbits. ⚖️🌍
- **(ii) The period of twilight and dawn increases polewards.**
    - **Answer:** Near the poles, the sun sets at a very shallow, slanting angle relative to the horizon. Because of this slow, diagonal descent, it takes a much longer time for the sun to reach 18° below the horizon (the end of twilight) compared to the steep, quick sunset at the Equator! 🌄📐
- **(iii) In New Zealand Christmas is celebrated in summer.**
    - **Answer:** New Zealand is in the Southern Hemisphere. In December, the Earth's position in its orbit means the Southern Hemisphere is tilted directly toward the Sun. Therefore, they experience their hot Summer season right when Christmas happens! 🎄🏖️

2. (d) Draw a well labelled diagram to show the tilt of the earth's axis and how the revolution causes seasons and variation in the length of day and night.

- **Answer:** *(To draw this 🖍️: Draw a Sun in the middle. Draw a large oval (ellipse) around it. Draw the Earth at the top, bottom, left, and right of the oval. Make sure the North Pole is leaning in the exact same direction (e.g., tilting right) on all four Earths. Shade the night half of each Earth and label the seasons for the Northern Hemisphere at each spot!)* 🪐🎨

V. Thinking Skills 🧠💡

1. Imagine the earth stops its rotation on its axis. What would be its impact on the inhabitants of the earth? Give examples to support your answer.

- **Answer:** It would be absolute chaos! 😱 The side facing the Sun would be permanently baked in scorching, never-ending daytime, drying up oceans. The side facing away would freeze into an eternal, icy night. The massive

temperature difference between the two sides would create violently extreme hurricane winds. Life would likely only survive in the narrow, permanent twilight zone between the burning hot and freezing cold halves! 🌪️🔥🧊

2. You travel for winter vacation to Australia from India. What type of climate will be there in Australia? Give reasons to support your answer.

- **Answer:** Pack your swimsuit, because you will experience a warm Summer climate! 🩳🏄 India is in the Northern Hemisphere (which is tilted away from the sun during winter). Australia is in the Southern Hemisphere. Due to Earth's permanent tilt, when it is winter in the North, the South is tilted directly *toward* the sun, basking in summer heat! 🌞AU

3. On June 21, your uncle who lives in New York experiences summer, whereas your cousin who lives in Jakarta, experiences winter. Give reasons to support your answer.

- **Answer:** On June 21 (Summer Solstice), the North Pole is physically leaning into the Sun. New York (in the Northern Hemisphere) gets highly concentrated, direct vertical rays, causing Summer heat. 🥵 Jakarta is located south of the Equator in the Southern Hemisphere, which is physically leaning *away* from the sun on that exact day, receiving weaker slanting rays and experiencing its cooler winter season! 🥶🗽

VI. Project/Activity (Section 2 text answers) 📊🎯

2. Draw a diagram to show the winter solstice and answer the following questions.

- **(i) Name the correct date depicted in your diagram.**
    - **Answer:** December 21st. 🗓️
- **(ii) Name the latitude where the sun is shining vertically overhead.**
    - **Answer:** Tropic of Capricorn ($23\frac{1}{2}^\circ$ S). 🎯
- **(iii) Name the part of the globe where there is night on this date.**
    - **Answer:** The entire region north of the Arctic Circle ($66\frac{1}{2}^\circ$ N to the North Pole) experiences 24-hour night. Also, the half of the globe currently facing away from the Sun. 🌌❄️

- **(iv) Name the latitude on which length of the day will be 24 hours.**
    - **Answer:** The Antarctic Circle (66½° S) and everywhere south of it to the South Pole. 🌞
- **(v) Give the date when the earth will be directly opposite to the above position.**
    - **Answer:** June 21st (Summer Solstice). 🔁
- **(vi) What will be the vertical position of the sun on that date?**
    - **Answer:** The sun will shine vertically overhead at the Tropic of Cancer (23½° N). 🌞
- **(vii) What is the effect of the inclination of the earth upon the length of day and night?**
    - **Answer:** It causes days to be extremely short and nights extremely long in the Northern Hemisphere, while causing days to be extremely long and nights short in the Southern Hemisphere. ⏳
- **(viii) What will be the probable length of day and night on that date?**
    - **Answer:** In the Northern Hemisphere (Winter), days are short (e.g., 9-10 hours) and nights are long (14-15 hours). In the Southern Hemisphere (Summer), days are long and nights are short. At the Equator, it remains exactly 12 hours of day and 12 hours of night! ⚖️🌓